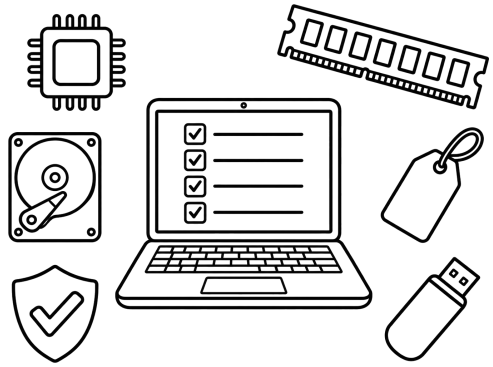
# ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

**GRADE 9**



# UNIT 1
# Preparation of Computer Specifications

Student Workbook


"The King of ICT"

## CHAMEEN RATHNASENA

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**STUDENT DETAILS**

**Name:** _____

**School:** _____

**Institution:** _____

**Contact:** _____

BEFORE WE START

# What This Unit Is About

Read this page with your teacher

## AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Name computer users and the work they do.
- [ ] Choose a type of computer to suit the user requirements.
- [ ] Sort peripheral devices into input, output and storage devices.
- [ ] Explain what specifications are and why they are important.
- [ ] Compare processors, hard disks, monitors and main memory.
- [ ] Explain the VGA, sound cards, ports and pre-installed software.
- [ ] Check warranty, price and after-sales services, and write a specification.

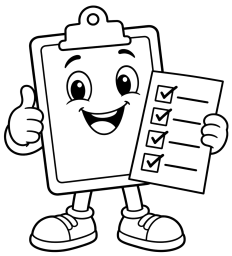
## THREE RULES FOR THE COMPUTER ROOM

- **Always** ask your teacher before you open a settings window on a lab computer.
- **Never** unplug a device or a cable while the computer is switched on.
- **Always** hold pen drives, memory cards and cables gently, and put them back.

## HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the activity: look at real computers, advertisements and price lists. Last you **write** the exercises in this book.

Colour the small pictures when you finish your work early.



Colour me!

## THE WORDS YOU WILL LEARN

| | | | |
|---|---|---|---|
| User | End user | User requirement | Workstation |
| All-in-one | Peripheral device | Specification | Processor |
| Gigahertz (GHz) | Dual core | Hard disk | Terabyte (TB) |
| Monitor size | Main memory | Gigabyte (GB) | VGA |
| HDMI | Sound card | Operating system | Open source |
| Warranty | After-sales service | USB port | RJ45 connector |

"A wise buyer writes the specification first and visits the shop second."

- THE KING OF ICT

CHAPTER 1

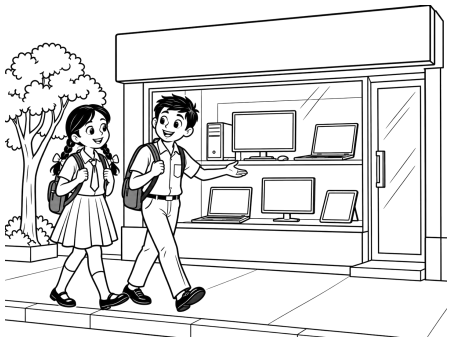
# Users, Computers and Peripheral Devices

**? AT THE COMPUTER SHOP**

"I need a computer for my school work and my father's business accounts."

"I need one I can carry, to show my graphic designs to my teacher."

Many types of computers are available. We can buy one, or get one **assembled** to suit our **requirements**.



## 1.1 Identifying the user

The one who uses a computer is called a **user**. People who work in different areas of ICT have different designations (Table 1.1).

*Table 1.1 - types of users and their work*

| User | Task |
|---|---|
| **Programmer** | Develops computer programs |
| **Network Administrator** | Manages and maintains computer networks |
| **System Analyst** | Designs information systems |
| **Software Engineer** | Develops software |
| **Computer Application Assistant** | Uses office application packages for office related tasks |
| **Web Developer** | Develops and maintains websites |

**Note:** users can be classified into two main categories: **system users** and **end users**. An end user uses the software maintained by the system user.

## 1.2 Selecting a computer to suit user requirements

**User requirements** are the tasks that are carried out by using a computer (Figure 1.1). A computer can be **selected** from those in the market (Figure 1.2), or it can be **assembled** to suit the user requirements.

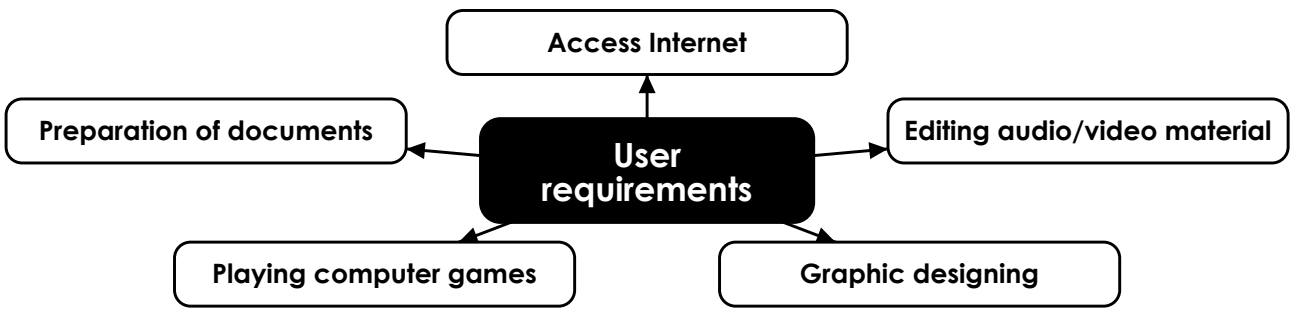



*Figure 1.1 - some examples of user requirements*

Computers can be classified according to their **nature and use**. **Non-portable computers** (servers, workstations, desktop and all-in-one computers) work with the **main electricity** power supply. They are large and relatively heavy, so they are installed in houses, schools or offices. **Computers for mobile use** (laptops, notebooks, tablets and smart phones) work with **rechargeable batteries**, so they can be used in buses, trains, aeroplanes or anywhere.

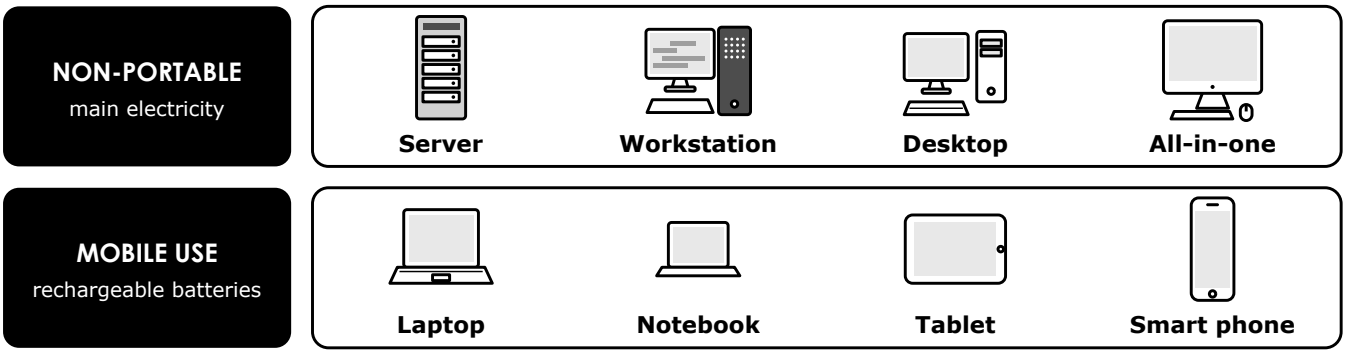



*Figure 1.2 - types of computers that are available in the market*

### DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 1.1)

Find the types of computers in your school and in advertisements.

| | |
|---|---|
| **1** | Walk round your school with your teacher: the computer lab, the office and the library. |
| **2** | Write the type of each computer you see (Figure 1.2): desktop, laptop, all-in-one ... |
| **3** | Write next to each one: non-portable or for mobile use. |
| **4** | Cut out computer advertisements from newspapers and sort them in the same way. |

### IMPORTANT - HOW TO LEARN MORE ABOUT COMPUTERS

- Advertisements (printed or electronic)
- Websites about computers
- Visiting computer shops
- Magazines and newspapers
- An expert in computers

"Every user is different. Choose the computer that fits your work, not your friend's."

- THE KING OF ICT



Colour me!

## 1.3 Computer peripheral devices

**Input devices** feed data and instructions into a computer. **Storage devices** store data. **Output devices** give the information processed from the input data. Input, storage and output devices are called **peripheral devices**.

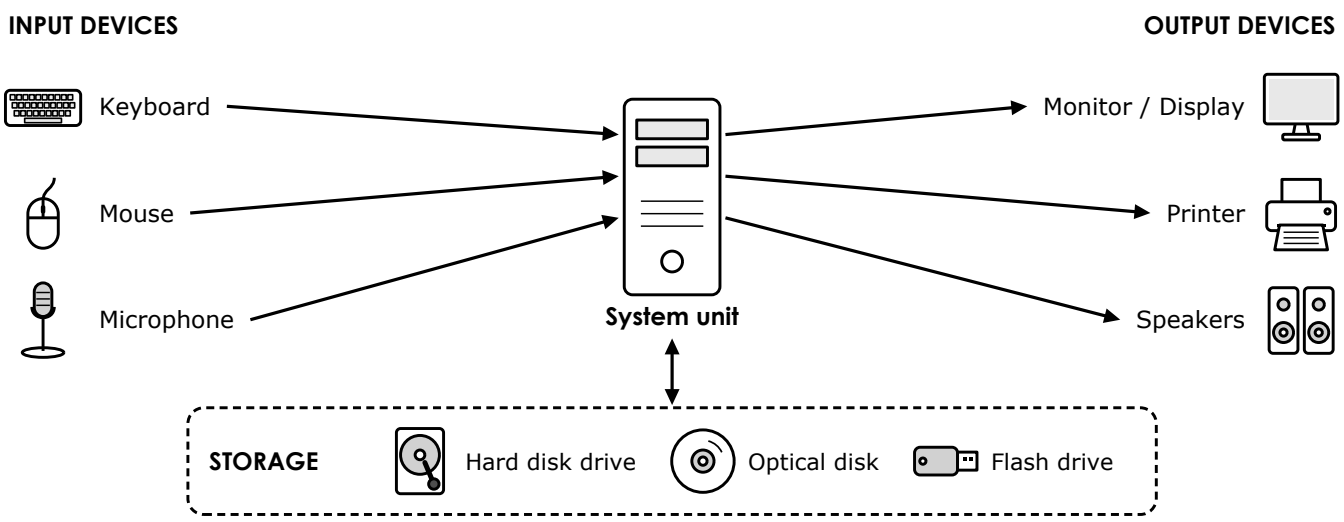



*Figure 1.3 - computer peripheral devices*

*Table 1.2 - peripheral devices*

| Input | Output | Storage |
|---|---|---|
| Keyboard | Monitor | Hard disk drive |
| Mouse | Printer | Optical disk drive |
| Microphone | Speaker | Flash drive |
| Touch screen | | Magnetic tape drive |

**NOTE**

A **touch screen** can input data and display information, so it is both an **input** and an **output** device.

**DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 1.2)**

Identify the peripheral devices of a computer in your laboratory.

| | |
|---|---|
| **1** | With your group, look at one computer in the lab. Do not unplug anything. |
| **2** | List every device that is connected to the computer. |
| **3** | Write I (input), O (output) or S (storage) next to each device (Table 1.2). |
| **4** | Is there a device that is both input and output? Name it. |

## A Match the users

**5 marks**

*Write the letter of the task in the box.*

| A - The user | | B - The task |
|---|---|---|
| 1. Programmer | | **A.** Designs information systems. |
| 2. Network Administrator | | **B.** Develops and maintains websites. |
| 3. System Analyst | | **C.** Develops computer programs. |
| 4. Computer Application Assistant | | **D.** Uses office packages for office related tasks. |
| 5. Web Developer | | **E.** Manages and maintains computer networks. |

## B Non-portable or mobile?

6 marks

*Tick one box for each computer.*

| The computer | Non-portable | Mobile use |
|---|---|---|
| Server computer | ☐ | ☐ |
| Laptop computer | ☐ | ☐ |
| All-in-one computer | ☐ | ☐ |
| Smart phone | ☐ | ☐ |
| Workstation | ☐ | ☐ |
| Tablet computer | ☐ | ☐ |

## C Input, output or storage?

8 marks

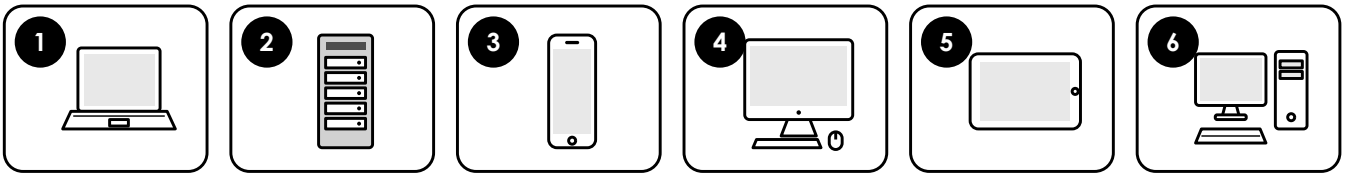
*Tick the right column. One device needs two ticks.*

| The device | Input | Output | Storage |
|---|---|---|---|
| Keyboard | ☐ | ☐ | ☐ |
| Monitor | ☐ | ☐ | ☐ |
| Flash drive | ☐ | ☐ | ☐ |
| Microphone | ☐ | ☐ | ☐ |
| Printer | ☐ | ☐ | ☐ |
| Hard disk drive | ☐ | ☐ | ☐ |
| Speaker | ☐ | ☐ | ☐ |
| Touch screen | ☐ | ☐ | ☐ |

## D Name the computers

6 marks

*Write the type of each computer on the line under it.*




---------- ---------- ---------- ---------- ---------- ----------

## E True or false

6 marks

*Write T or F in the box.*

1. ☐ A Web Developer develops and maintains websites.
2. ☐ Users can be classified into system users and end users.
3. ☐ Server computers are computers for mobile use.
4. ☐ Computers for mobile use can be used in buses, trains and aeroplanes.
5. ☐ A printer is an input device.
6. ☐ A touch screen is both an input device and an output device.

CHAPTER 2

# Specifications and the Processor

**? TWO BOOKS, TWO JOBS**

"I want an exercise book for maths and a drawing book for art, please."

"Square ruled or single ruled? How many pages? Big or small?"

The answers to the shopkeeper's questions are the **specifications** of a book. A computer has specifications too, and the most important one is its **processor**.



## 1.4 Computer specifications

Before we buy an item, we must know its **value** and its **quality**. **Specifications** are the common features of an item. The length, the width and the thickness of the paper decide the quality of an exercise book.

*The specifications of three books (textbook pages 6 - 7)*

| Specification | Exercise book | Maths exercise book | Art drawing book |
|---|---|---|---|
| **Length** | 210 mm | 210 mm | 300 mm |
| **Width** | 148 mm | 148 mm | 210 mm |
| **Number of pages** | 40 | 200 | 20 |
| **Thickness of paper** | 60 GSM | 60 GSM | 70 GSM |
| **Type** | Single ruled | Square ruled | Blank |

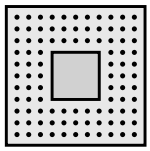
Different purposes need different specifications, and the prices are different too. In the same way, the specifications of a computer depend on its **use**.

## 1.5 Specifications of computers and peripheral devices

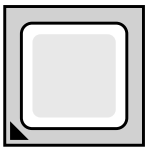
The important factors of the specification of a computer are its processor, hard disk, monitor, main memory, VGA, sound card and pre-installed software. **One can generally assure the quality of an item through its specifications.**

## 1.5.1 The processor

The **brain** helps a human to make decisions and to respond swiftly. In the same way the **processor** (Central Processing Unit) is the most important part of a computer: it processes data swiftly. So the processor is called the **"brain" of the computer**.



*The side that connects to the motherboard*

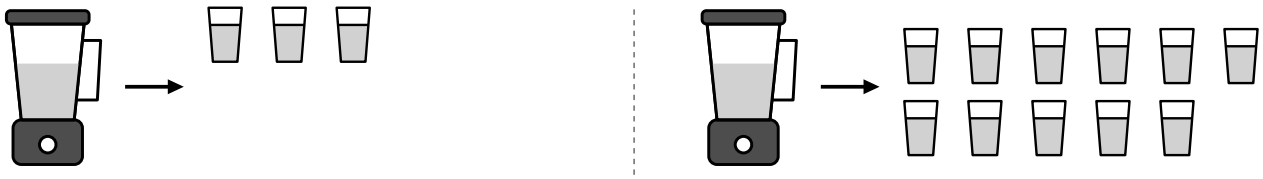


*View from above*

*Figure 1.5 - the Central Processing Unit*

## Speed of the processor

A slower machine takes a longer time to do a task; a faster machine does more work in a unit time. In Figure 1.6 the faster blender makes much more juice in the same time.



**Slower speed: 3 glasses** **Higher speed: 11 glasses**

*Figure 1.6 - blenders working at different speeds*

A processor with a **high speed** processes more data in a unit time, so the computer performs better and software runs faster. Select a processor with a **greater speed**.

**! IMPORTANT**

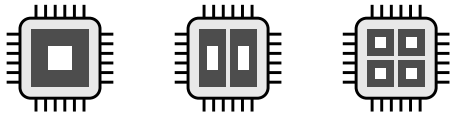
The speed of a processor is the **number of instructions executed per second**. It is measured in **Hertz**: megahertz (**MHz**) or gigahertz (**GHz**). **1000 MHz = 1 GHz**

## Processor manufacturers and types

**Apple**, **Intel** and **AMD** (Advanced Micro Devices) are processor manufacturers. A CPU is classified by the **number of processors** in it (Table 1.3). When this number increases, its **capacity** also increases.

*Tables 1.3 and 1.4 - types of processors*

| No. of CPUs | Type | Intel examples |
|---|---|---|
| 1 | Single Core | Pentium I, II, III, IV |
| 2 | Dual Core | Dual Core / Core 2 Duo |
| 4 | Quad Core | Core i3, i5, i7, i9 |



**Single Core** **Dual Core** **Quad Core**

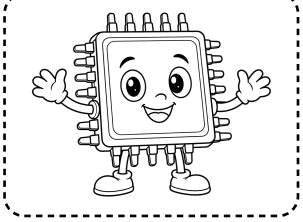
**> DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 1.3)**

Find the processor of a lab computer and its speed.

| | |
|---|---|
| **1** | With your teacher, right-click **This PC** and click **Properties** (or Settings - System - About). |
| **2** | Find the line **Processor**. Write the name of the processor and its speed. |
| **3** | Is the speed in MHz or in GHz? Write it in both units (1000 MHz = 1 GHz). |
| **4** | Use Table 1.4: is it a single core, a dual core or a quad core processor? |

"The processor is the brain of the computer. Keep your own brain running at full speed."

- THE KING OF ICT



Colour me!

## A Fill in the blanks

8 marks

*Use the words in the box.*

**WORD BANK**

| | | | |
|---|---|---|---|
| quality | specifications | brain | data |
| instructions | Hertz | AMD | capacity |

1. Before we buy an item, we must know its value and its ..... .
2. The common features of an item, such as length and width, are its ..... .
3. The processor is called the ..... of the computer.
4. A processor with a high speed processes more ..... in a unit time.
5. The speed of a processor is the number of ..... executed per second.
6. The speed of a processor is measured in ..... (MHz or GHz).
7. Apple, Intel and ..... are processor manufacturers.
8. When the number of processors in a CPU increases, its ..... also increases.

## B MHz and GHz

6 marks

*Remember: 1000 MHz = 1 GHz.*

| | |
|---|---|
| **1.** 2 GHz = ..... MHz | **4.** 3000 MHz = ..... GHz |
| **2.** 2.8 GHz = ..... MHz | **5.** 3200 MHz = ..... GHz |
| **3.** 3.4 GHz = ..... MHz | **6.** 500 MHz = ..... GHz |

## C Which processor is faster?

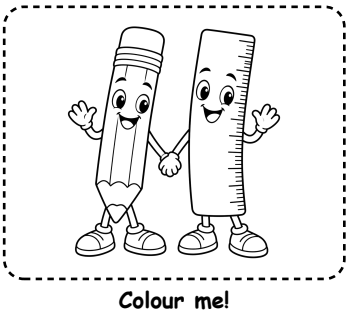
5 marks

*Tick the faster processor in each row.*

| | Processor A | Processor B |
|---|---|---|
| **1.** | ☐ 2.8 GHz | ☐ 3.4 GHz |
| **2.** | ☐ 3.2 GHz | ☐ 3000 MHz |
| **3.** | ☐ 2800 MHz | ☐ 3.0 GHz |
| **4.** | ☐ 3400 MHz | ☐ 3.2 GHz |
| **5.** | ☐ 1 GHz | ☐ 900 MHz |

"A faster processor finishes the work sooner. So does a focused student."

- THE KING OF ICT



Colour me!

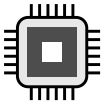
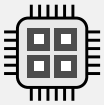
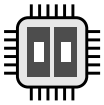
## D Complete the processor table

9 marks

*Count the processors in each chip. Use the words in the box.*

**WORD BANK**

Single Core  Dual Core  Quad Core  Pentium IV  Core 2 Duo  Core i7

| The chip | No. of CPUs | Type | Name (Intel) |
|---|---|---|---|
|  | | | |
|  | | | |
|  | | | |

## E True or false

6 marks

*Write T or F in the box.*

1. ☐ Specifications help us to know the quality of an item.
2. ☐ A maths exercise book and an art drawing book have the same specifications.
3. ☐ A machine working at a higher speed does more work in a unit time.
4. ☐ 1 GHz is equal to 100 MHz.
5. ☐ A dual core processor has two processors in it.
6. ☐ Core i3, i5, i7 and i9 are single core processors.

"Two cores work better than one. Two friends learn better than one."

- THE KING OF ICT

CHAPTER 3

# Hard Disk, Monitor, Memory and Software

### ? NO SPACE, NO SPEED

"My computer says the disk is full, and my new game is so slow!"

"Then look at its hard disk, its main memory and its VGA."

In this chapter you learn the specifications of the **hard disk**, the **monitor**, the **main memory**, the **VGA**, the **sound card** and the **software**.



## 1.5.2 Hard disk

The **hard disk** gives **permanent** storage space to store data and to install all the software. It is the main **secondary storage** device of a computer. A 160 page exercise book gives more writing space than a 40 page book. In the same way, a hard disk with a greater **capacity** stores more data.

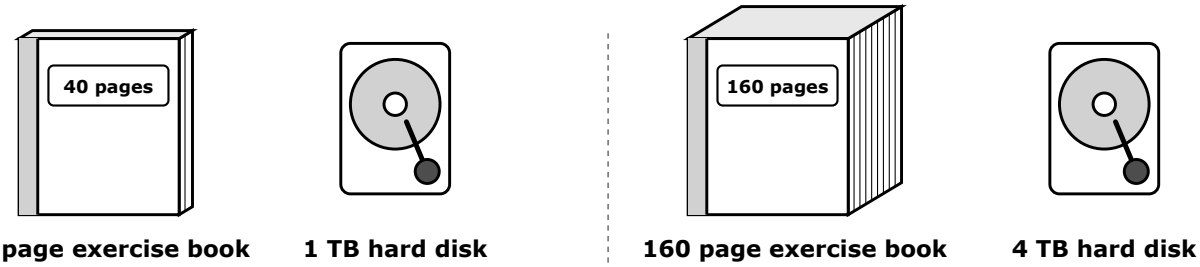



*Figure 1.8 - an analogy for hard disk capacities*

### DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 1.4)

Find the capacity of the hard disk of a lab computer.

| | |
|---|---|
| **1** | Open **File Explorer** and click **This PC**. |
| **2** | Look at **Local Disk (C:)**. Write its total size and its free space. |
| **3** | Write the size of any other drive (D:, E:) too. |
| **4** | Compare with your friends: which computer can store the most data? |

## 1.5.3 The monitor display

The **monitor** is the main output device of a computer. Most often a user works with a computer through its monitor.

**Monitor size** - a wall clock is larger than a wrist watch, so its time is easy to see. In the same way, a **larger monitor** has better visibility and is useful to view a larger picture. The size of a monitor is the length of its **diagonal**, measured in **inches** (Figure 1.9).

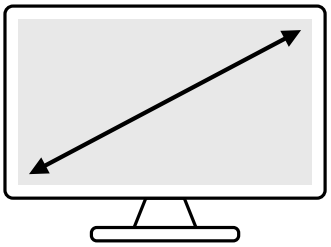



*Figure 1.9 - monitor size*

**Monitor technology** - there are three types of monitors:

- **CRT** - Cathode Ray Tube monitor
- **LCD** - Liquid Crystal Display monitor
- **LED** - Light Emitting Diode monitor

CRT monitors use **more electricity**, are **heavier** and take more space. Light LCD and LED monitors use less electricity and are common today.

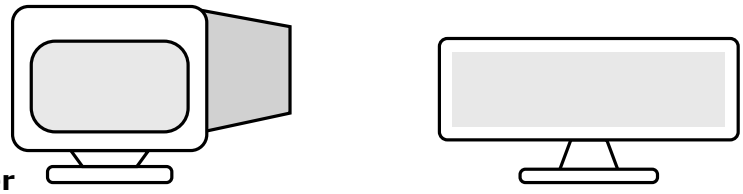



*Figure 1.10 - CRT and LCD / LED monitors*

## 1.5.4 Main memory

In a library, students keep their school bags in a rack and take them back when they leave. In the same way, data and instructions are stored **temporarily** in the **main memory** while the computer works. A bigger rack holds more bags: a main memory with a **higher capacity** stores more data and instructions, so it is better. A **RAM** (Random Access Memory) card acts as the main memory.

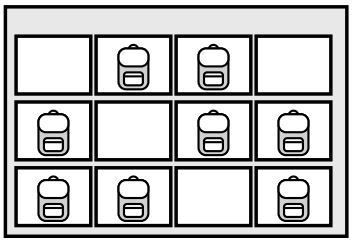


*Figure 1.11 - a rack for school bags*

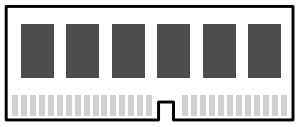


*Figure 1.12 - a RAM card*

**IMPORTANT**

The capacity of the main memory is measured in **bytes**: megabytes (MB) or gigabytes (GB).
**1024 MB = 1 GB**

**DO IT - PRACTICAL 5 (WORKBOOK ACTIVITY 1.5)**

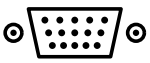
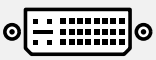
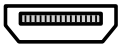
Find the capacity of the main memory (RAM) of a lab computer.

| | |
|---|---|
| **1** | Open the **About** page again (This PC - Properties). |
| **2** | Find **Installed RAM** and write the capacity of the main memory. |
| **3** | Write it in MB too (1 GB = 1024 MB). |
| **4** | Which computer in your lab has the biggest main memory? |

## 1.5.5 Video Graphic Adapter (VGA)

The output is fed to the monitor through the **Video Graphic Adapter (VGA)**. An **on board VGA** is fixed to the motherboard. A **separate VGA card** can be fixed to the motherboard; it has its own **video memory** and **processor**, so it is useful for playing computer games. Modern computers use **DVI** or **HDMI** ports instead of the VGA port (Figure 1.14).

*Figure 1.14 - types of video ports*

| Port | Name | Full name | Remember |
|---|---|---|---|
|  | **VGA** | Video Graphic Array / Adapter | Only images; sound needs a separate cable. |
|  | **DVI** | Digital Visual Interface | Used in modern computers instead of VGA. |
|  | **HDMI** | High Definition Multimedia Interface | Sound and video in one cable. |

"Store your lessons on your hard disk, not in RAM. Revise them, and they stay with you."

- THE KING OF ICT

## 1.5.6 Sound cards

A **sound card** is needed to listen to music and to record audio. A **microphone** on the sound card records (inputs) sound; a **speaker** plays (outputs) sound. Most computers have the sound card **built into the motherboard**; an **external** sound card can be connected if necessary. The ports use a standard colour scheme:

### THE COLOURS OF THE SOUND PORTS

| Port | Colour | Name | Connect |
|---|---|---|---|
| | **Light green** | Audio out | a speaker or headphones |
| | **Light pink** | Mic in | a microphone |
| | **Light blue** | Line in | sound from external devices |

## 1.5.7 Pre-installed software

The **operating system** provides the interface between the user and the hardware. It is essential to run application software, which is installed after the operating system (Figure 1.17).

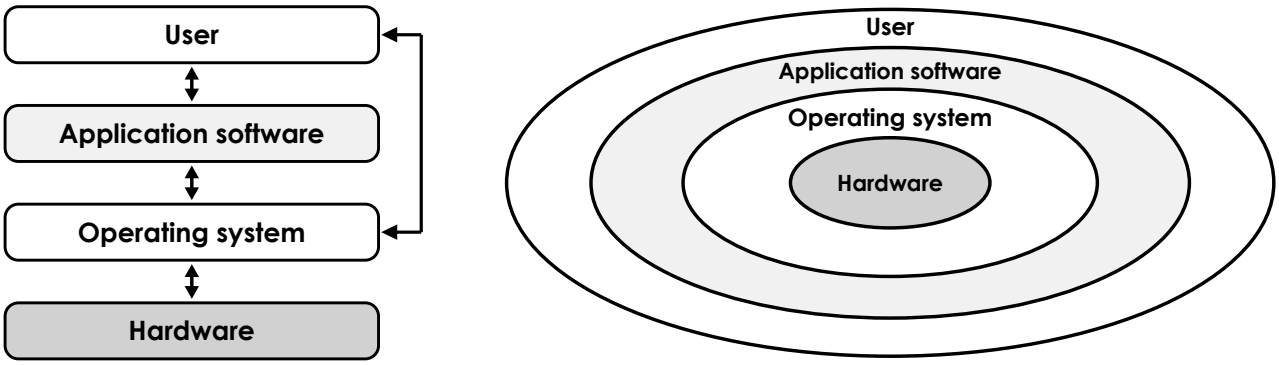



*Figure 1.17 - hardware, software and user*

| Operating system | Examples | Remember |
|---|---|---|
| **Free and open source** | Linux, Ubuntu | Obtained free of charge. |
| **Licensed** | Windows 8, Windows 10 | Must be purchased. The computer costs more. |
| **Trial version** | A trial version of Windows 10 | Can be used only for a limited period. |

When you buy a desktop or a laptop, check whether an operating system is **pre-installed**. Most often only DOS or Linux is pre-installed. The application software you need depends on your requirements.

### A MB and GB

6 marks

*Remember: 1024 MB = 1 GB.*

| | |
|---|---|
| **1.** 1 GB = ..... MB | **4.** 8 GB = ..... MB |
| **2.** 2 GB = ..... MB | **5.** 2048 MB = ..... GB |
| **3.** 4 GB = ..... MB | **6.** 512 MB = ..... GB |

## B Match the examples

6 marks

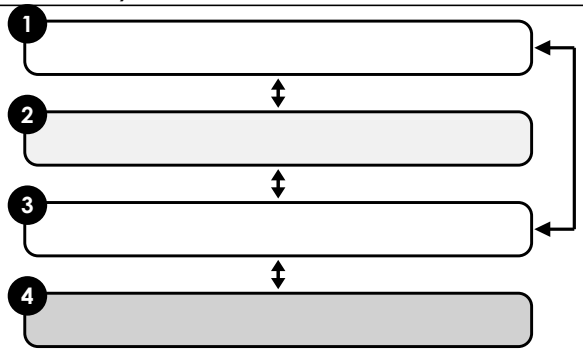
*Each example in the textbook explains a part of the computer.*

| A - The example | | B - In a computer |
|---|---|---|
| 1. The brain of a human | | **A.** A larger monitor |
| 2. A blender working at a higher speed | | **B.** The specifications of a computer |
| 3. A 160 page exercise book | | **C.** The processor of a computer |
| 4. A wall clock that is easy to see | | **D.** The main memory (RAM) |
| 5. A rack for school bags in a library | | **E.** A hard disk with a greater capacity |
| 6. The length, width and pages of a book | | **F.** A processor with a greater speed |

## C Complete Figure 1.17

4 marks

*Write the layers in the numbered boxes.*




**USE THESE WORDS**

Hardware
User
Operating system
Application software

## D VGA, DVI or HDMI?

5 marks

*Tick one box for each sentence.*

| The port ... | VGA | DVI | HDMI |
|---|---|---|---|
| It sends both sound and video signals. | ☐ | ☐ | ☐ |
| It sends only images, not sound. | ☐ | ☐ | ☐ |
| Digital Visual Interface | ☐ | ☐ | ☐ |
| High Definition Multimedia Interface | ☐ | ☐ | ☐ |
| Video Graphic Array (or Adapter) | ☐ | ☐ | ☐ |

## E Which sound port?

4 marks

*Write the colour of the port you would use.*

| You want to connect ... | Colour of the port |
|---|---|
| Speakers | |
| A microphone | |
| Headphones | |
| Sound from an external device (line in) | |

## F Free, licensed or trial?

6 marks

*Tick one box for each operating system or sentence.*

| The operating system | Free and open source | Licensed | Trial version |
|---|---|---|---|
| Linux | ☐ | ☐ | ☐ |
| Windows 10 bought from a shop | ☐ | ☐ | ☐ |
| Ubuntu | ☐ | ☐ | ☐ |
| Windows 8 | ☐ | ☐ | ☐ |
| It can be used only for a limited period. | ☐ | ☐ | ☐ |
| It can be obtained free of charge. | ☐ | ☐ | ☐ |

## G True or false

6 marks

*Write T or F in the box.*

1. ☐ The hard disk stores data and software permanently.
2. ☐ A 1 TB hard disk can store more data than a 4 TB hard disk.
3. ☐ The size of a monitor is measured from its left side to its right side.
4. ☐ CRT monitors use less electricity than LCD and LED monitors.
5. ☐ The main memory stores data and instructions temporarily.
6. ☐ A separate VGA card has its own video memory and processor.

CHAPTER 4

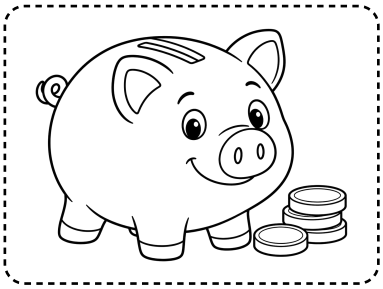
# Buying the Right Computer

### ? THE CHEAPEST ONE?

"Two shops sell the same laptop. One is cheaper. Let's buy it!"

"Wait! The other shop gives a longer warranty and a helpdesk."

Price is not everything. In this chapter you learn the **non-technical** things to check - warranty, price and after-sales services - and the **ports** a computer needs.



Colour me!

## 1.6 Non-technical features to consider

When we buy an exercise book, we also think of the manufacturer, the type of book and the price. In the same way, **non-technical** specifications must be considered when we buy a computer.

## 1.6.1 Warranty

The **warranty** given by the vendor and the manufacturer covers the defects of a product for a certain period. There are three kinds of warranty:

| Badge | Kind of warranty | What it means |
|---|---|---|
|   | **a. Manufacturer warranty** | Given by the company that made the computer. If the device does not work properly during the warranty period, it is repaired or replaced with a new one. |
|   | **b. Extended warranty** | A longer warranty given **in addition to** the standard warranty. The customer pays an **additional cost** for it. |
|   | **c. On-site warranty** | Technicians from the company **visit the place of work** to repair the product. They maintain it, replace faulty parts and examine how the machine works. |

Every computer has a standard warranty, but some parts may have a different warranty. For example, a laptop with a 3 year warranty may have only a 1 year warranty for its battery.

## 1.6.2 Price

The price of a computer depends on the customer's requirements. For example:

1. A computer with a **high speed processor** is rather high in price.
2. A computer with a **large monitor** is comparatively high in price.
3. A computer with a **lesser memory capacity** is low in price.
4. A computer with a **high capacity hard disk** is high in price.

Compare the prices of computers that suit your requirements at different places, and select the one with the lowest price.

**! IMPORTANT**

It is **not** advisable to buy a computer by considering **only its price**.

**> DO IT - PRACTICAL 6 (WORKBOOK ACTIVITIES 1.6 AND 1.7)**

How do specifications change the price? Where is the price lowest?

| | |
|---|---|
| **1** | Collect the advertisements of two computers from newspapers or websites. |
| **2** | Copy their processor speed, memory, hard disk, monitor size and price. |
| **3** | Underline what is higher in the more expensive computer. Why does it cost more? |
| **4** | Now find the price of one computer at three places. Which price is the lowest? |
| **5** | Is the lowest price also the best choice? Check the warranty too. |

## 1.6.3 Services after-sale

Computer sales centres give these **after-sales services** to their customers:

1. Technical advice
2. Technical assistance
3. Telephone, e-mail and website information (contact information) of the vendor

First **prepare your specification**. Then buy the computer from a vendor with a **reputed name**, made by a **recognized manufacturer**.

### DO IT - PRACTICAL 7 (WORKBOOK ACTIVITY 1.8)

Find the after-sales services of computer vendors.

| | |
|---|---|
| **1** | Look at the advertisements or the websites of two computer vendors. |
| **2** | Find the after-sales services they give: technical advice and technical assistance. |
| **3** | Write down their telephone number, e-mail address and website. |
| **4** | Which vendor would you buy from? Give a reason. |

## 1.6.4 Ports and network connections

**Ports** connect the peripheral devices to a computer. Choose peripheral devices to suit your requirements - but the computer must have the **ports** to connect them.



**USB port**

**a. Universal Serial Bus (USB)** - the USB port is the **most widely used** port to connect peripheral devices. So a computer must have **several USB ports**.

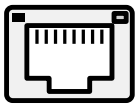
### DEVICES THAT CAN BE CONNECTED THROUGH A USB PORT

- Printer
- External hard disk
- Bar code reader
- Keyboard
- External DVD drive
- Memory card
- Mouse
- Digital camera
- Pen drive
- Scanner
- Web camera

### DO IT - PRACTICAL 8 (WORKBOOK ACTIVITY 1.9)

Find the USB ports of a lab computer and the devices that use them.

| | |
|---|---|
| **1** | Count the USB ports at the front and at the back of a lab computer. |
| **2** | List the devices that are connected through USB ports now. |
| **3** | Name three more devices from the USB list that you could connect. |
| **4** | With your teacher, plug a pen drive into a USB port and open it in File Explorer. |



Network port

**b. Network port / RJ45 connector** - a computer uses the **RJ45 connector** to connect to a **network** (Figure 1.18).





**c. Bluetooth and Wi-Fi** - they give **cable free** (wireless) network connections to computers.

## Basic specifications to consider when you buy a computer

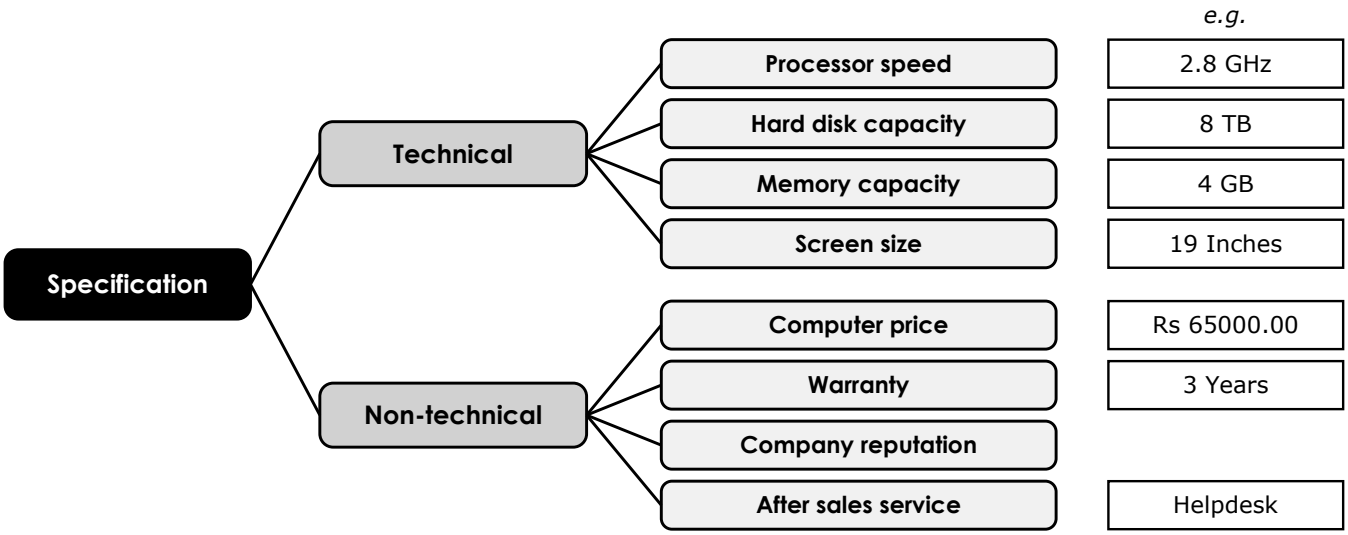



*Figure 1.19 - technical and non-technical specifications*

### DO IT - PRACTICAL 9 (WORKBOOK ACTIVITY 1.10)

Sort the specifications of a computer advertisement.

| | |
|---|---|
| **1** | Take a computer advertisement from a newspaper or a website. |
| **2** | Draw a tree like Figure 1.19 in your exercise book. |
| **3** | Write each specification in the technical branch or in the non-technical branch. |
| **4** | Write the example value next to each one, for example 2.8 GHz or 3 years. |

## Important factors of technical specifications

| Part | Specification | Examples |
|---|---|---|
| **Central Processing Unit** | Type | Single Core / Dual Core / Quad Core |
| | Speed | 2.8 / 3.0 / 3.2 / 3.4 GHz |
| **Main memory** | Capacity | 512 MB / 1 GB / 2 GB / 4 GB / 8 GB |
| | Generation | $1^{st}$, $2^{nd}$, $3^{rd}$, $4^{th}$ etc. |
| **Hard disk** | Capacity | 500 GB, 750 GB, 1 TB, 2 TB, 4 TB |
| **Monitor** | Size | 17 inch, 19 inch, 21 inch |
| | Type | CRT / LCD / LED |
| **Video Graphic Adapter** | Type | VGA / DVI / HDMI |
| **Sound card** | Type | Onboard, Separate |

## DO IT - PRACTICAL 10 (WORKBOOK ACTIVITY 1.11)

Prepare the specification of a computer for a user.

| | |
|---|---|
| **1** | Choose a user: a student, a graphic designer, a shop owner or a gamer. |
| **2** | Write the user requirements of that user (Figure 1.1). |
| **3** | Choose each technical specification from the table on page 20 of the textbook. |
| **4** | Add the non-technical factors: warranty, price and after-sales services. |
| **5** | Present your specification to the class and explain your choices. |

## A Which warranty is it?

6 marks

*Tick one box for each sentence.*

| The warranty ... | Manufacturer | Extended | On-site |
|---|---|---|---|
| It is given by the manufacturing company. | ☐ | ☐ | ☐ |
| Given in addition to the standard warranty. | ☐ | ☐ | ☐ |
| Technicians come to your office to repair it. | ☐ | ☐ | ☐ |
| The customer pays an additional cost for it. | ☐ | ☐ | ☐ |
| A faulty device is replaced with a new one. | ☐ | ☐ | ☐ |
| Faulty parts are replaced at your workplace. | ☐ | ☐ | ☐ |

## B Technical or non-technical?

8 marks

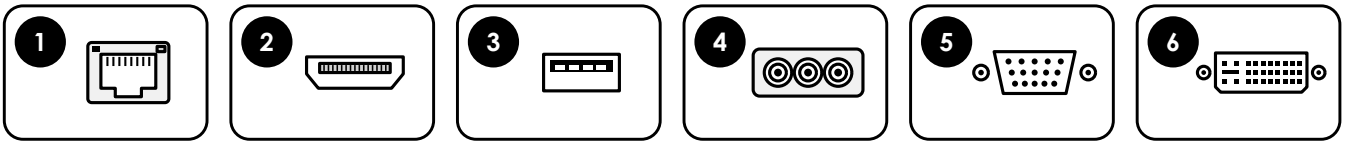
*Tick one box for each specification.*

| The specification | Technical | Non-technical |
|---|---|---|
| Processor speed | ☐ | ☐ |
| Warranty | ☐ | ☐ |
| Memory capacity | ☐ | ☐ |
| Company reputation | ☐ | ☐ |
| Screen size | ☐ | ☐ |
| After sales service | ☐ | ☐ |
| Hard disk capacity | ☐ | ☐ |
| Computer price | ☐ | ☐ |

## C Name the ports

6 marks

*Write the name of each port on the line under it.*



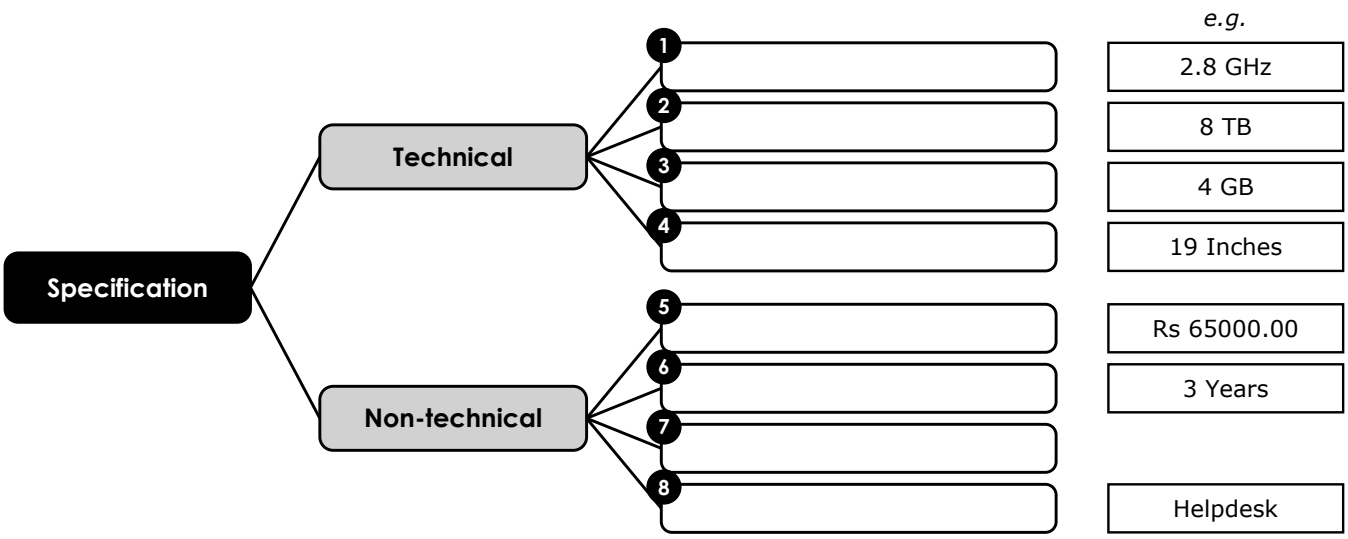
## D Complete Figure 1.19

**8 marks**

*Write the specifications in the numbered boxes. The examples help you.*

**WORD BANK**

| | | | |
|---|---|---|---|
| After sales service | Company reputation | Computer price | Hard disk capacity |
| Memory capacity | Processor speed | Screen size | Warranty |




## E Compare two computers

**6 marks**

*Read the two advertisements and answer.*

| COMPUTER A | |
|---|---|
| **Processor** | Dual Core, 2.8 GHz |
| **Main memory** | 4 GB |
| **Hard disk** | 500 GB |
| **Monitor** | 19 inch LED |
| **VGA** | On board |
| **Operating system** | Linux |
| **Warranty** | 1 year |
| **Price** | Rs 65000.00 |

| COMPUTER B | |
|---|---|
| **Processor** | Quad Core, 3.4 GHz |
| **Main memory** | 8 GB |
| **Hard disk** | 2 TB |
| **Monitor** | 21 inch LED |
| **VGA** | Separate VGA card |
| **Operating system** | Windows 10 (licensed) |
| **Warranty** | 3 years on-site |
| **Price** | Rs 120000.00 |

| Question | Your answer |
|---|---|
| 1. Which computer has the faster processor? (1 mark) | |
| 2. Which computer can store more data? (1 mark) | |
| 3. Which computer is better for playing computer games? Why? (1 mark) | |
| 4. Which computer has a free and open source operating system? (1 mark) | |
| 5. Write two reasons why Computer B costs more. (2 marks) | |

REVISION

# Unit 1 in One Page

Read this the night before the test

| Topic | The key facts |
|---|---|
| **Users** | Programmer, Network Administrator, System Analyst, Software Engineer, Web Developer ... System users and end users. |
| **User requirements** | Access Internet, documents, editing audio/video, graphic designing, computer games. |
| **Types of computers** | Non-portable: server, workstation, desktop, all-in-one. Mobile use: laptop, notebook, tablet, smart phone. |
| **Peripheral devices** | Input: keyboard, mouse, microphone. Output: monitor, printer, speaker. Storage: hard disk, optical disk, flash drive. |
| **Specifications** | The common features of an item; they show its quality. Different uses need different specifications. |
| **Processor** | The brain of the computer. Speed in Hertz: 1000 MHz = 1 GHz. Single, dual, quad core (1, 2, 4 processors). |
| **Hard disk** | Permanent storage; the main secondary storage. 4 TB stores more than 1 TB. |
| **Monitor** | The main output device. Size: the diagonal in inches. CRT uses more electricity than LCD, LED. |
| **Main memory** | The RAM card stores data and instructions temporarily. 1024 MB = 1 GB. |
| **VGA and sound** | Separate VGA card for games. HDMI: sound and video. Green: audio out, pink: mic in, blue: line in. |
| **Software** | The operating system: the interface between the user and the hardware. Free, licensed, trial. |
| **Non-technical** | Warranty (manufacturer, extended, on-site), price, after-sales services, reputation. |
| **Ports** | USB: the most widely used port. RJ45: network port. Bluetooth and Wi-Fi: wireless. |

## ★ TICK WHEN YOU CAN DO IT

- ☐ I can name the users of computers and the work they do.
- ☐ I can sort computers into non-portable and mobile computers.
- ☐ I can sort peripheral devices into input, output and storage.
- ☐ I can explain what specifications are and why they are important.
- ☐ I can compare processors, hard disks, monitors and main memory.
- ☐ I can explain warranty, price and after-sales services.
- ☐ I can prepare a computer specification to suit a user.

WORD POWER

# Unit 1 Word List

Twenty-eight words you must know

| Word | What it means |
|---|---|
| **User** | The one who uses a computer. |
| **End user** | A user who uses the software maintained by the system user. |
| **User requirement** | A task that is carried out by using a computer. |
| **Non-portable computer** | A large computer that works with the main electricity: server, desktop ... |
| **Mobile computer** | A computer that works with rechargeable batteries: laptop, tablet ... |
| **Peripheral device** | An input, storage or output device of a computer. |
| **Specification** | A detailed description of an item, about its quality or quantity. |
| **Processor (CPU)** | The most important part of a computer: its "brain". |
| **Hertz (Hz)** | The unit of processor speed: 1000 MHz = 1 GHz. |
| **Single / dual / quad core** | A CPU with 1, 2 or 4 processors in it. |
| **Hard disk** | The main secondary storage: it keeps data and software permanently. |
| **Monitor size** | The length of the diagonal of the screen, in inches. |
| **CRT, LCD, LED** | Cathode Ray Tube, Liquid Crystal Display, Light Emitting Diode monitors. |
| **Main memory (RAM)** | Stores data and instructions temporarily while the computer works. |
| **Megabyte, gigabyte** | Units of memory capacity: 1024 MB = 1 GB. |
| **VGA** | Video Graphic Adapter: it feeds the output to the monitor. |
| **DVI, HDMI** | Digital Visual Interface, High Definition Multimedia Interface (sound and video). |
| **Sound card** | Needed to listen to music and to record audio. |
| **Operating system** | The interface between the user and the hardware. |
| **Free and open source** | Software that can be obtained free of charge (Linux, Ubuntu). |
| **Licensed software** | Software that must be purchased (Windows 8, Windows 10). |
| **Trial version** | Software that can be used only for a limited period. |
| **Warranty** | A promise to cover the defects of a product for a certain period. |
| **On-site warranty** | Technicians visit the place of work to repair the product. |
| **After-sales services** | Technical advice, technical assistance and contact information. |
| **USB** | Universal Serial Bus: the most widely used port for peripheral devices. |
| **RJ45 connector** | It connects a computer to a network. |
| **Bluetooth, Wi-Fi** | Cable free (wireless) network connections. |

"A word you understand is a tool you can use. Learn them all."

- THE KING OF ICT

END OF UNIT

# Unit 1 Test

Time: 45 minutes Total: 40 marks

**Name** _____ **Class** _____ **Marks** ____

## 1 Choose the correct answer

**8 marks**

*Underline your answer.*

1. Who designs information systems?
   (a) a programmer (b) a web developer
   (c) a system analyst (d) an end user
2. Which of these is a computer for mobile use?
   (a) Server (b) Workstation (c) All-in-one (d) Tablet
3. Which of these is a storage device?
   (a) Microphone (b) Flash drive (c) Speaker (d) Monitor
4. The speed of a processor is measured in
   (a) inches (b) gigabytes (c) gigahertz (d) GSM
5. How many processors are there in a quad core CPU?
   (a) 1 (b) 2 (c) 3 (d) 4
6. Which port sends both sound and video?
   (a) VGA (b) HDMI (c) RJ45 (d) USB
7. A microphone is connected to the sound port coloured
   (a) light green (b) light blue (c) light pink (d) black
8. A - CRT monitors use more electricity than LED monitors. B - RAM stores data permanently. C - Ubuntu is free and open source. Which are correct?
   (a) A and B only (b) A and C only (c) B and C only (d) A, B and C

## 2 Fill in the blanks

**8 marks**

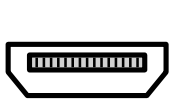
*One word or number in each blank.*

1. Input, storage and output devices are called ..... devices.
2. The processor is called the ..... of the computer.
3. 1 GHz = ..... MHz
4. 1 GB = ..... MB
5. The size of a monitor is measured diagonally in ..... .
6. The hard disk is the main ..... storage device of a computer.
7. A warranty given in addition to the standard warranty is an ..... warranty.
8. A computer uses an ..... connector to connect to a network.

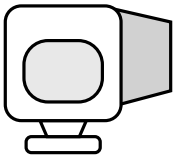
## 3 Look at the pictures

8 marks

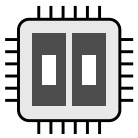
*Write the name of each one, and what it does or one fact about it.*

1 

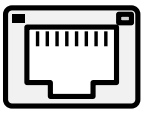
Name: .....

2 

Name: .....

3 

Name: .....

4 

Name: .....

## 4 Answer these

16 marks

*Write in the space given.*

1. Name the two groups of computers according to their nature and use. Give two examples of each. (3 marks)

2. Why is it advisable to select a processor with a greater speed? (2 marks)

3. Name three technical specifications and three non-technical specifications of a computer. (3 marks)

4. Explain the three kinds of warranty. (3 marks)

5. A projector is connected with an HDMI cable, and another one with a VGA cable. What is the difference? (2 marks)

6. Name the three after-sales services given by computer sales centres. (3 marks)